Morgan Burke

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INDIVIDUAL SUMMARY

Recent graduate in BSc Software Engineering, achieving a First Class with Honours at Cardiff University, now based in Brisbane on a Working Holiday visa with eligibility for two further years without the need for 88 days of specified work. A full-stack web developer working mainly in React, TypeScript, Node.js and Python, with experience building AI-driven web applications and leading a 30+ member autonomous racing team sponsored by IBM. Looking for a full-time or contract role where I can take on meaningful engineering work, learn from experienced teams and keep developing as an engineer.

TECHNICAL SKILLS

Full-Stack Web Based Applications Developer
Languages: React, Typescript, JavaScript, Python, HTML, CSS, C++, Kotlin
Frameworks: React, Express.js, Node.js, Spring, Express
DevOps/Tools: Git, GitLab, GitHub, Docker, Linux, Jenkins, Git CI/CD, Git Actions
Testing: Junit, Jest, test plans
Databases: MySQL, MariaDB, MongoDB, SQL
Concepts: REST APIs, OOP, Agile, Waterfall, CI/CD, Code Reviews, Peer Programming

PROJECTS

Artificial Intelligence Triaging System – Software Engineering Dissertation Project
React, TypeScript, Python, Node.js, Express.js, MongoDB, Vite, HTML5, CSS, Jest, JWT, SpaCy NER

- Designed and implemented a full stack web-based application with integrated microservices which triages users to specialist clinicians from patient free text symptom descriptions.
- Designed and developed a Python FASTAPI microservice and created an Orchestrator layer which processed user input.
- Created a validated user intake form which collected patient demographic data.
- Contributed across the frontend, backend and microservices, implementing role-based access and session management utilising Microsoft Azure, Microsoft Authentication library and Microsoft's Entra ID.
- Designed, developed, trained and implemented a bespoke Name Entity Recognition (NER) algorithm paired with a normalisation pipeline to extract symptoms from free text.

Cardiff Autonomous Racing System – Team Principal
Python, C++, YOLOV8, Docker

- Led development of an autonomous racing software stack, integrating perception, planning and control modules.
- Led the implementation of computer vision pipelines with YOLOv8 to detect cones and create bounding boxes in real time.
- Procured and collaborated with Industry sponsors such as IBM and Delaware UK, having presented technical progress at IBM TechXChange 2025, Orlando FL.

NFL Fan Android App – Independent Project
Kotlin, Android Studio, REST APIs

- Developed a native android app providing NFL statistics, team data and player stats using a public API.
- Designed and developed intuitive front-end UI, implemented data caching and custom theming for improved UX.
- Followed MVVM architecture with ViewModels for maintainable and testable code.

Wii Portfolio Site – Independent Project
HTML, CSS, JavaScript, QuickTime, GitHub Pages

- Designed and built a personal portfolio website inspired by the Nintendo Wii Interface.
- Implemented interactive menus and transitions using JavaScript and embedded multimedia via QuickTime elements.
- Focused on creating a fun, intuitive experience for users whilst ensuring a responsive design and accessibility.

EXPERIENCES

Cainan, Freelance Software Engineer; July 2026 - Present:

- I founded Cainan, my freelance software development company.
- Website development, Software Solutions and Contract Software Engineering Work
- Designed, developed and deployed carylcomedy.com, cainan.studio and gwib.co.uk

IBM TechXChange 2025, Invited Guest Speaker; Oct 2025:

- Invited speaker at 8,000+ attendee IBM TechXChange 2025 in Orlando Florida.
- Delivered a session on the work of Cardiff Autonomous Racing, highlighting our progress with support from IBM and other industry partners, and outlining how the team develops students' technical and professional skills.

Cardiff Autonomous Racing (CAR), Team Principal & Founder; Sep 2024 – Jul 2026:

- Founded and led a 30+ Student team with Cardiff University's School of Computer Science and Informatics.
- Secured funding and partnerships with major industry sponsors, including IBM, Delaware UK&I and BIOS IT.
- Directed project planning, budgeting and technical development of an autonomous racing system.
- Mentored team members and fostered interdisciplinary communication amongst team members.
- Related Media & Recognition: IBM Personal Article, Cardiff University Article, Delaware UK Article.

Tesco DotCom Department, Customer Delivery Driver; May 2024 – Present

Yoichi Eco Village, Hokkaido Japan, Volunteer; Aug - Sep 2024:

- Supported community-led environmental and sustainability project.
- Collaborated with an international team developing cross-cultural communication skills and participating in cultural exchange.
- Contributed to hands-on tasks on the farm, demonstrating adaptability and initiative in a new environment differing from my studies.

Novomatic Gaming UK, Test Administrator; Jul – Sep 2023 & Jun – Aug 2024 (Contract)**:**

- Conducted extensive manual software testing and quality assurance on gaming systems intended for the gambling industry.
- Documented and reported test results, ensuring compliance with internal standards.
- Coordinated with developers to identify and troubleshoot system issues.

EDUCATION

BSc Applied Software Engineering, Cardiff University, School of Computer Science and Informatics; 2023 – 2026:

- First Class Honours
- *Additional Module:* Beginners Japanese (A1) Achieved: 2:1

Bryntirion Comprehensive School, Bridgend; 2016 – 2023:

- A-Levels: Computer Science B, History B, Welsh Baccalaureate B

ADDITIONAL INFORMATION

Languages: English (native), Welsh, (beginner), Japanese (beginner).
Interests: American Football, Tabletop Gaming, Rugby Union, Video Games, Surfing, Motorcycles, Medieval & Modern History and Alternative Music
Societies: Cardiff University American Football Team, Alternative Music Society
Citizenship: British, Irish
Drivers Licenses: Full UK Driver's License (Clean – 0 Points)